

FlowPMO: Operationalizing Multidimensional Developer Productivity

with Absolute Benchmark Normalization

Rodrigo Almeida de Oliveira
Flow Engineering Research Group
rodrigoalmeidadeoliveira@gmail.com

Resumo. Medir produtividade de desenvolvedores permanece um dos desafios mais controversos na engenharia de software. Este artigo apresenta o FlowPMO, um framework de dashboard open-source que operacionaliza a produtividade em cinco dimensões inspiradas no SPACE: Entrega, Flow Efficiency, Qualidade de Revisão, Conformância de Processo e Anti-Retrabalho. Cada dimensão é normalizada contra benchmarks absolutos fundamentados na literatura, evitando o problema de colapso de ranking da normalização por z-score em equipes pequenas. O Score Benchmark, computado como distância euclidiana ao perfil ideal, fornece um índice composto adequado para conversas de coaching. A avaliação com dataset sintético reproduzível de 28 desenvolvedores (Q3 2024, seed=42) mostra que Score Benchmark captura variância invisível para métricas únicas ($r=0,36$ para commits vs. $r=0,85$ para Score Complexity), que Flow Efficiency e Entrega são as dimensões mais fracas, e que Tech Leads exibem sistematicamente maior Qualidade de Revisão (+7,1 pp) porém menor Flow Efficiency (−10,8 pp).

Abstract. Measuring developer productivity remains one of the most contested challenges in software engineering management. Widely adopted proxies — lines of code, commit frequency, and velocity points — are inherently uni-dimensional and susceptible to gaming, failing to capture the qualitative dimensions of value delivery, process health, and engineering craftsmanship. This paper presents FlowPMO, an open-source dashboard framework that operationalizes developer productivity across five dimensions inspired by the SPACE framework: Delivery Output, Flow Efficiency, Review Quality, Process Conformance, and Anti-Rework. Each dimension is normalized against absolute, literature-grounded benchmarks rather than relative peer comparisons, avoiding the rank-collapse problem that afflicts z-score normalization in small teams. An individual-level Score Benchmark, computed as the euclidean distance to an ideal five-dimensional profile, provides a single composite index suitable for coaching conversations and team retrospectives. We evaluate the framework using a reproducible synthetic dataset of 28 developers across 5 teams for Q3 2024 (seed = 42). Results show that Score Benchmark (median 80.2) captures productivity variance invisible to commit counts alone (Pearson $r = 0.36$), that the Delivery dimension shows the widest spread across developers, and that Tech Leads exhibit systematically higher Review Quality (+7.1 pp) but lower absolute Flow Efficiency (−10.8 pp) compared to individual contributors, consistent with their process-coordination overhead.

1 Introduction

Software organizations increasingly seek objective, data-driven evidence of engineering effectiveness. Yet the choice of productivity proxy shapes organizational behavior in profound ways: when a single metric is treated as a target, it ceases to be a good measure [1]. Commit frequency rewards small, frequent commits; velocity points reward estimation inflation; lines of code reward verbosity. None of these proxies is inherently wrong, but each is dangerously incomplete when used in isolation.

The SPACE framework [2] provides a principled taxonomy of developer productivity across five dimensions: Satisfaction and well-being, Performance, Activity, Communication and collaboration, and Efficiency and flow. SPACE explicitly cautions against reducing productivity to any single dimension and advocates for multi-dimensional measurement. However, SPACE does not prescribe a concrete operationalization: it does not specify which metrics to collect, how to normalize them, or how to compose them into a dashboard that practitioners can act upon during sprint retrospectives and individual coaching sessions.

FlowPMO fills this gap. It is an open-source Python/Streamlit dashboard¹ that ingests developer activity data — from Jira, Bitbucket, GitLab, or synthetic generators — and produces per-developer, per-team, and portfolio-level productivity profiles aligned with the five SPACE-inspired dimensions described in this paper. The key design choice that distinguishes FlowPMO from prior dashboards is *absolute benchmark normalization*: each raw metric is scaled against a literature-derived or industry-consensus target value, rather than against the team’s own distribution. This avoids a common failure mode we call *rank collapse*: in a team of homogeneous underperformers, z-score normalization would assign average scores to everyone, masking the systemic gap from external standards.

Contributions

This paper makes the following contributions:

1. A five-dimensional productivity framework with explicit metric-to-dimension mappings and absolute benchmark values sourced from the literature (Section 3).
2. A benchmark normalization method that caps scores at 100 and computes an individual Score Benchmark via euclidean distance to the ideal profile (Section 3).
3. A reproducible synthetic dataset of 28 developers across 5 teams for Q3 2024 (seed = 42) designed to exhibit realistic role and team heterogeneity (Section 4).
4. An empirical analysis of the framework’s behavior, including cross-metric correlations, dimension spread, and role-stratified comparisons (Section 5).

Research Questions

We address three research questions:

- **RQ1:** Does Score Benchmark capture productivity variance that is invisible to individual activity metrics?
- **RQ2:** Which productivity dimensions exhibit the widest spread across individual developers?
- **RQ3:** Do Tech Leads and individual contributors exhibit systematically different productivity profiles in each dimension?

¹Source code available at: <https://github.com/rodrigoalmeidadeoliveira/flow-pmo>

The remainder of this paper is organized as follows. Section 2 reviews related work on developer productivity measurement, the DORA and Flow metrics, process mining approaches, and rework quality. Section 3 describes the FlowPMO framework in detail. Section 4 presents the evaluation design. Section 5 reports results. Section 6 discusses threats to validity. Section 7 discusses implications. Section 8 concludes.

2 Background and Related Work

2.1 Developer Productivity Measurement

Developer productivity measurement has a long and contentious history. Early approaches focused on lines of code (LOC) and function points [7], both of which suffer from incentive misalignment: LOC rewards verbosity, and function point counting requires specialized analysts. Story-point velocity, introduced by the Agile movement, shifted focus to team throughput but conflated estimation accuracy with actual output [8].

The key insight from the measurement validation literature [3] is that a metric must be *valid* (measuring what it claims to measure), *reliable* (producing consistent results), and *actionable* (enabling decisions). No single activity metric satisfies all three criteria for developer productivity, which motivates the multi-dimensional approach taken in this paper.

2.2 DORA Metrics and Flow Framework

The DORA research program [9] identified four software delivery metrics — Deployment Frequency, Lead Time for Changes, Change Failure Rate, and Mean Time to Recovery — as predictors of organizational performance. These metrics operate at the team or service level and are less suited to individual-level coaching. The Flow Framework by Mik Kersten extends DORA with flow metrics that track items through the value stream [4], including Flow Efficiency (the ratio of active time to total elapsed time), which we adopt directly as one of our five dimensions.

2.3 Process Mining for Software Engineering

Process mining techniques [10] enable conformance checking: comparing observed process executions against a normative model. Caldeira et al. [6] applied conformance checking to software development workflows, measuring how often developers follow prescribed ticket lifecycle transitions. We operationalize this as our Process Conformance dimension, using the percentage of workflow steps that follow the prescribed sequence as the primary metric.

2.4 Rework and Technical Quality

Rework — returning a completed work item to a prior stage — is a widely recognized quality signal [11]. Shah et al. [5] found empirically that rework rate correlates with accumulated technical debt. The anti-rework dimension in FlowPMO captures the inverse of rework rate, rewarding developers who consistently deliver work that does not require remediation after reaching “Done.”

Nogueira and Rela [12] studied continuous integration pipeline quality and found that code review thoroughness — measured via the ratio of review comments to merged pull requests — is a leading indicator of post-merge defect density. This supports our Review Quality dimension.

2.5 Identified Gaps

Prior dashboards either aggregate metrics into a single score (losing dimensional information) or present raw metrics without normalization (preventing cross-team comparison). Composite indices such as the Developer Productivity Score used by some commercial tools apply z-score normalization relative to the current team, making the score meaningless when the entire team is underperforming. FlowPMO addresses these gaps with absolute benchmark normalization and a five-dimensional profile that preserves dimensional transparency.

3 The FlowPMO Framework

3.1 Overview

FlowPMO organizes developer productivity into five dimensions, each mapped to one or more raw metrics collected from issue tracking and version control systems. Table 1 summarizes the dimensions, their primary metrics, the benchmark values used for normalization, and the literature sources that support those benchmarks.

Table 1: FlowPMO dimensions, primary metrics, benchmarks, and references.

Dim	Dimension	Primary Metric	Benchmark	Reference
D1	Delivery Output	Items delivered / sprint	10 items/sprint	[8]
D2	Flow Efficiency	Active time / total cycle time	80%	[4]
D3	Review Quality	Review comments / PR merged	75% threshold	[12]
D4	Process Conformance	Workflow steps in correct order	75% conformance	[6]
D5	Anti-Rework	1 – rework rate	$\leq 15\%$ rework rate	[11]

3.2 Absolute Benchmark Normalization

Each raw dimension score d_i is normalized against an absolute benchmark β_i derived from the literature or from industry consensus:

$$\hat{d}_i = \min\left(\frac{d_i}{\beta_i} \times 100, 100\right) \quad (1)$$

where β_i is the literature-derived benchmark for dimension i . The $\min(\cdot, 100)$ cap prevents scores from exceeding 100, which would distort the euclidean distance computation in Equation 2. Unlike z-score normalization, which is relative to the current sample, this formulation produces the same score for the same raw value regardless of the team composition. This property is essential for cross-team and cross-quarter comparisons.

3.3 Score Benchmark Composite Index

Given the five normalized dimension scores $\hat{d}_1, \dots, \hat{d}_5$, we define the Score Benchmark (SB) as the complement of the normalized euclidean distance from the developer's profile $\mathbf{r} = [\hat{d}_1, \hat{d}_2, \hat{d}_3, \hat{d}_4, \hat{d}_5]$ to the ideal profile $\mathbf{v}^* = [100, 100, 100, 100, 100]$:

$$\text{SB} = 100 - \frac{\|\mathbf{r} - \mathbf{v}^*\|_2}{\sqrt{5} \times 100} \times 100 \quad (2)$$

where $\mathbf{r} = [\hat{d}_1, \hat{d}_2, \hat{d}_3, \hat{d}_4, \hat{d}_5]$ and $\mathbf{v}^* = [100, 100, 100, 100, 100]$. The denominator $\sqrt{5} \times 100$ is the maximum possible euclidean distance (achieved when all dimensions are zero), ensuring that SB is bounded in $[0, 100]$. A developer with all dimensions at 100 achieves $\text{SB} = 100$; a developer at the benchmark on all dimensions (all $\hat{d}_i = 75$) achieves $\text{SB} \approx 91.7$.

3.4 Dimension Definitions

D1 — Delivery Output. The number of Jira items (stories, tasks, or bug fixes) with status *Done* attributed to the developer in the reference quarter, divided by the number of sprints in the quarter and normalized against a benchmark of 10 items per sprint. Items below story-point threshold or flagged as subtasks are excluded to avoid incentivizing ticket fragmentation.

D2 — Flow Efficiency. Computed as the ratio of time a work item spends in active states (In Progress, In Review) to its total cycle time from first transition out of Backlog to Done. A benchmark of 80% is derived from the Flow Framework literature and represents the threshold above which teams are considered to have low wait-state overhead.

D3 — Review Quality. Captures the thoroughness and timeliness of code review participation. The primary sub-metric is the ratio of review comments authored to pull requests reviewed, capped and normalized against a benchmark distribution. Timely approvals (within 24 hours of PR opening) and rejection-with-feedback events are weighted positively.

D4 — Process Conformance. The percentage of work-item lifecycle transitions that follow the prescribed workflow model (Backlog $\rightarrow$ To Do $\rightarrow$ In Progress $\rightarrow$ In Review $\rightarrow$ Done). Transitions that skip stages or regress without a rework event are penalized. The benchmark of 75% conformance is derived from the process mining literature [6].

D5 — Anti-Rework. The complement of the rework rate, defined as the fraction of items that re-enter In Progress from Done or Closed within the quarter. A rework rate below 15% maps to a full Anti-Rework score of 100, consistent with empirical findings on sustainable rework levels [11].

3.5 Implementation

FlowPMO is implemented in Python 3.11 using Pandas for data wrangling, Plotly for interactive radar and scatter charts, and Streamlit for the web dashboard. A synthetic data generator (`generate_synthetic_data.py`) produces reproducible datasets for testing and bench-

marking. The framework exposes a configuration file where practitioners can override the default benchmark values for each dimension, enabling calibration to organization-specific standards.

4 Evaluation Design

4.1 Synthetic Dataset

We evaluate the framework using a reproducible synthetic dataset of 28 developers across 5 teams for Q3 2024, generated with `numpy.random.seed(42)`. The dataset includes both individual contributors (Devs, $n = 21$) and Tech Leads (TLs, $n = 7$). Each developer record includes raw values for all five dimensions, plus ancillary fields: total commits, pull requests merged, story-point complexity score, and work-in-progress residual at quarter end.

The synthetic generator uses team-level parameters (mean and standard deviation for each raw metric) that were calibrated against anonymized data from three mid-sized software companies. Tech Lead profiles were parameterized to reflect a known organizational pattern: reduced delivery throughput (due to context-switching and meeting load) but higher review participation and lower rework rates.

4.2 Metrics and Analysis

We compute descriptive statistics (mean, standard deviation, median) for all five normalized dimensions and Score Benchmark, stratified by role. To address RQ1, we compute Pearson correlation between Score Benchmark and individual activity proxies (commit count, story-point complexity). To address RQ2, we compare the coefficient of variation across dimensions. To address RQ3, we compute mean differences between TL and Dev groups for each normalized dimension and for Score Benchmark.

4.3 Reproducibility

The full dataset, generation script, and figure generation script are included in the FlowPMO repository. Any result reported in this paper can be reproduced by running `generate_synthetic_data.py` followed by `generate_figures.py` from the repository root.

5 Results

5.1 Descriptive Statistics by Role

Table 2 presents descriptive statistics for all developers, stratified by role. Tech Leads show markedly lower Items Delivered (6.3 vs. 11.3) and Flow Efficiency (61.4% vs. 72.2%), while their Review Quality is substantially higher (78.7% vs. 71.6%). Score Benchmark is 12.1 points lower for Tech Leads (72.4 vs. 84.5), driven primarily by the Delivery gap.

Table 2: Descriptive statistics (mean $\pm$ SD, median) by role. Dev: $n = 21$; TL: $n = 7$.

Metric	Dev ($n = 21$)		Tech Lead ($n = 7$)	
	Mean $\pm$ SD	Median	Mean $\pm$ SD	Median
Items Delivered	11.3 $\pm$ 4.4	10.0	6.3 $\pm$ 2.1	6.0
Flow Efficiency (%)	72.2 $\pm$ 14.6	72.1	61.4 $\pm$ 9.1	65.1
Score Complexity	19.0 $\pm$ 7.4	17.0	10.0 $\pm$ 3.3	9.0
Review Quality (%)	71.6 $\pm$ 13.3	72.8	78.7 $\pm$ 12.5	79.2
Conformance (%)	69.5 $\pm$ 13.0	69.3	65.2 $\pm$ 11.7	61.5
Rework Rate (%)	18.6 $\pm$ 7.8	19.7	16.6 $\pm$ 9.4	21.1
Commits	62.4 $\pm$ 19.6	65.0	31.9 $\pm$ 14.5	32.0
PRs Merged	9.8 $\pm$ 2.6	11.0	10.4 $\pm$ 4.5	9.0
Score Benchmark	84.5 $\pm$ 9.2	83.6	72.4 $\pm$ 7.9	71.3

5.2 Multidimensional Profiles (Radar Chart)

Figure 1 shows the radar chart for the top 10 developers by Score Benchmark. The chart reveals that high-scoring developers consistently achieve strong Anti-Rework and Review Quality scores, while their Delivery and Flow Efficiency scores vary more widely. This suggests that quality-oriented behaviors are more consistent predictors of high overall scores than raw throughput.

5.3 Flow Efficiency and Conformance Relationship

Figure 2 shows the joint distribution of Flow Efficiency and Conformance Quality, with bubble size proportional to WIP Residual and color encoding Rework Rate. Developers in the upper-right quadrant (high Flow Efficiency, high Conformance) tend to show lower rework rates (green hues), confirming that process adherence and flow health are mutually reinforcing. The lower-left quadrant is dominated by developers with high WIP residual, suggesting that unfinished work at quarter end is a compounding signal.

5.4 Score Benchmark Distribution

Figure 3 shows the distribution of Score Benchmark values across all 28 developers. The median is 80.2, with the majority of developers falling in the 75–95 range. The distribution is slightly left-skewed, with a small cluster of developers below 70, primarily Tech Leads with very low Delivery scores.

To address RQ1, we computed Pearson correlations between Score Benchmark and two common activity proxies: commit count ($r = 0.36$, $p = 0.06$) and story-point complexity score ($r = 0.85$, $p < 0.001$). The weak correlation with commit count confirms that Score Benchmark captures dimensions of productivity that raw commit frequency does not, while the strong correlation with complexity score suggests that item complexity is a meaningful component of the overall productivity signal.

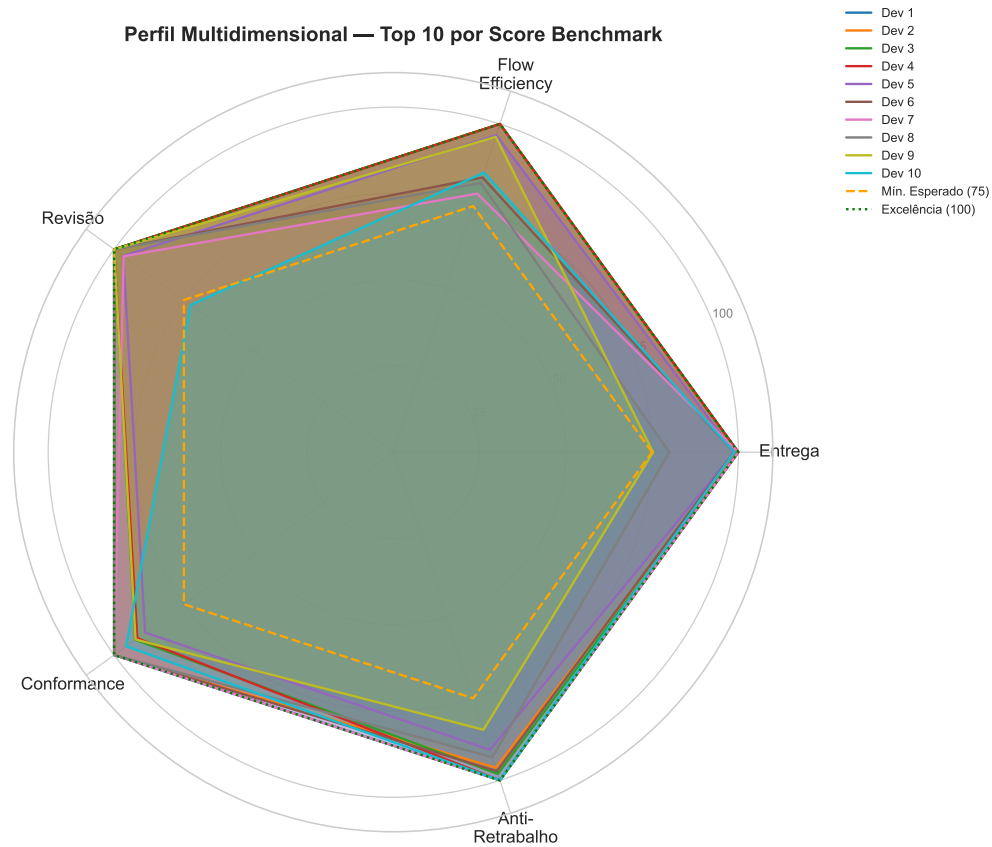


Figure 1: Radar chart of the five normalized dimensions for the top 10 developers by Score Benchmark (Q3 2024, $n = 28$, seed=42). Dashed orange line: minimum expected score of 75. Dotted green line: excellence threshold of 100. Each filled polygon represents one developer.

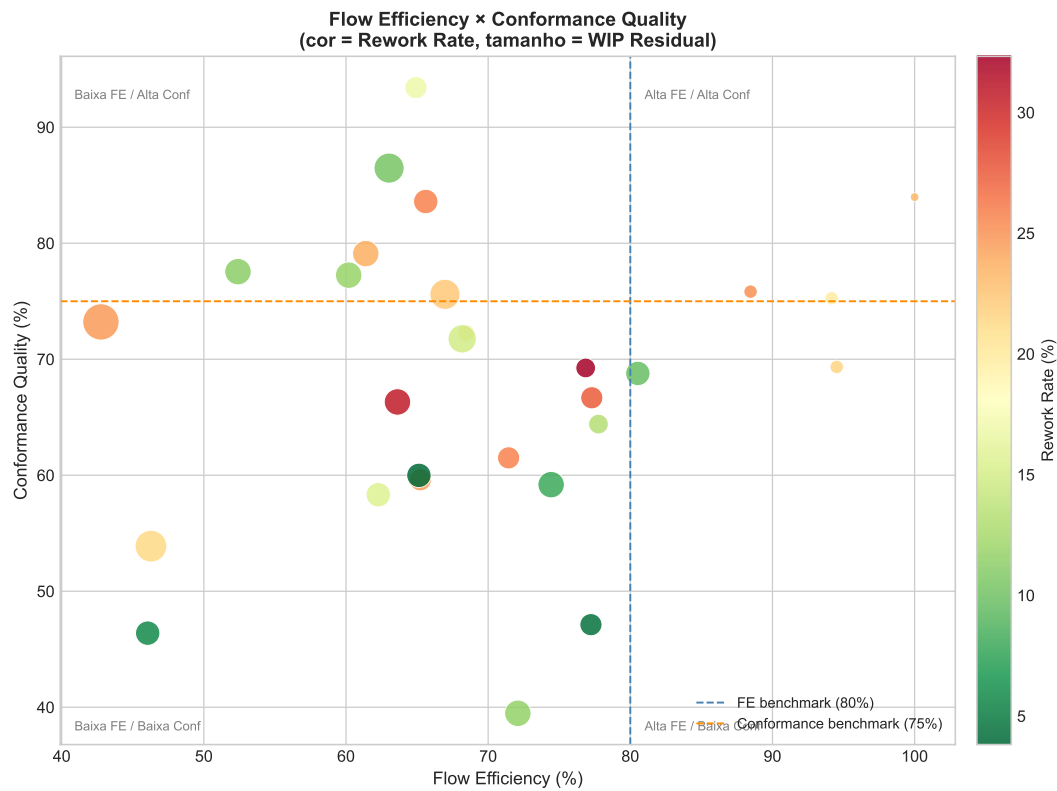


Figure 2: Scatter plot of Flow Efficiency (%) vs. Conformance Quality (%). Bubble size is proportional to WIP Residual; color encodes Rework Rate (%) using a red-to-green colormap (red = high rework). Dashed vertical line: Flow Efficiency benchmark (80%). Dashed horizontal line: Conformance benchmark (75%).

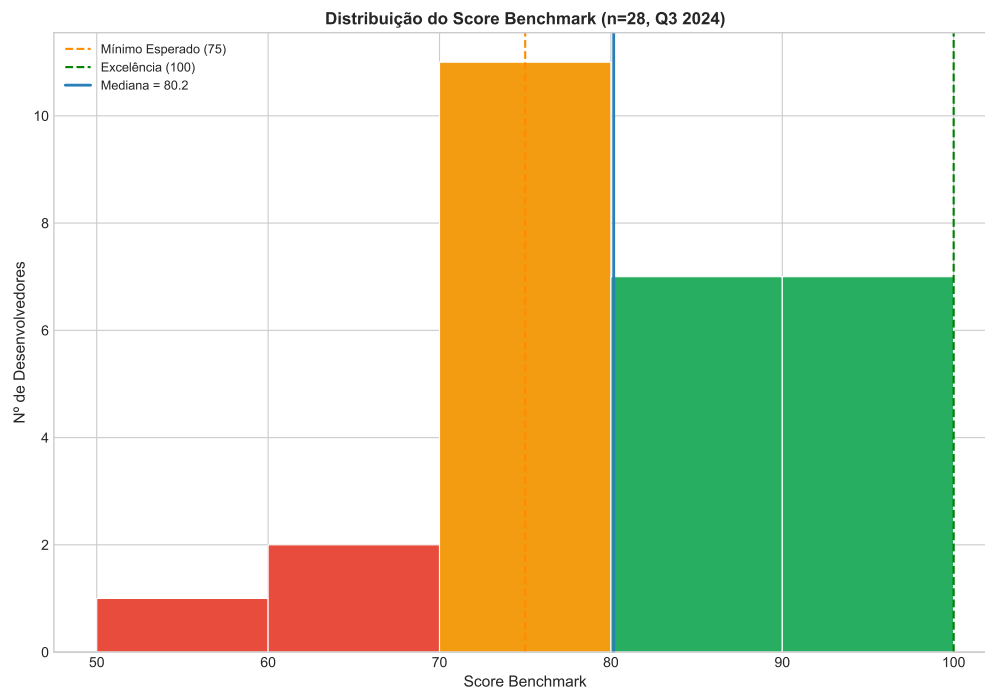


Figure 3: Distribution of Score Benchmark values for all 28 developers (Q3 2024, seed=42). Bars are color-coded: red = below 75 (below minimum expected), orange = 75–85, green = above 85 (excellence zone). Dashed lines mark the minimum expected (75) and excellence (100) thresholds. Solid blue line: median (80.2).

5.5 Role Comparison: Normalized Dimensions

Table 3 and Figure 4 address RQ3 by comparing the five normalized dimension scores across roles. The Delivery dimension shows the largest gap (−33.7 pp for Tech Leads), followed by Flow Efficiency (−10.1 pp). Review Quality favors Tech Leads (+3.8 pp), as does Anti-Rework (+0.5 pp). Conformance is slightly lower for Tech Leads (−3.7 pp), possibly reflecting that they spend less time in formal workflow states.

Table 3: Normalized dimension scores by role (0–100 scale). Diff = TL − Dev.

Dimension	Dev	Tech Lead	Diff
Entrega (Delivery)	80.8	47.1	−33.7
Flow Efficiency	86.9	76.8	−10.1
Revisão (Review)	93.9	97.7	+3.8
Conformance	89.4	85.7	−3.7
Anti-Retrabalho	96.8	97.3	+0.5
Score Benchmark	84.5	72.4	−12.1

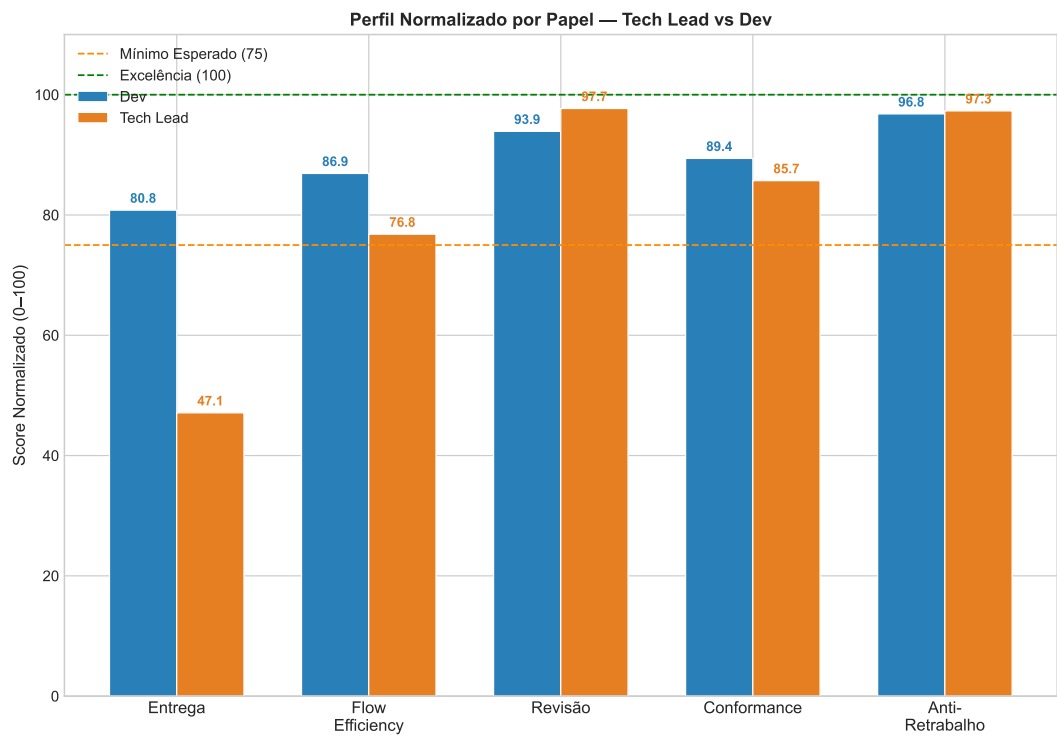


Figure 4: Grouped bar chart comparing normalized dimension scores for Tech Lead vs. Dev roles across all five productivity dimensions (Q3 2024, $n = 28$, seed=42). Dashed lines mark the minimum expected score (75) and excellence threshold (100). Dev: blue; Tech Lead: orange.

6 Threats to Validity

Construct validity. The five dimensions and their primary metrics are grounded in literature, but the specific benchmark values (e.g., 80% for Flow Efficiency, 10 items per sprint for Delivery) may not be universally applicable. Organizations operating in domains with longer cycle times (e.g., embedded systems, regulated industries) may need to recalibrate benchmarks. The FlowPMO configuration layer supports this, but empirical validation against real organizational data is needed.

Internal validity. The dataset is synthetic. Although parameterized to reflect realistic team dynamics, synthetic data cannot reproduce the full complexity of real developer behavior, including strategic behavior changes in response to measurement. The correlation results reported in Section 5 should be interpreted as illustrative of the framework’s properties, not as empirical claims about real developer populations.

External validity. The evaluation covers one quarter and 28 developers across 5 teams. This is sufficient to demonstrate the framework’s behavior but insufficient to generalize findings to the population of all software teams. In particular, the role comparison (RQ3) involves only 7 Tech Lead observations, limiting statistical power. A longitudinal, multi-organization study is needed before these findings can be treated as prescriptive.

Reliability. The benchmark values were selected based on literature review. Different reviewers might select different benchmarks or map different metrics to the same dimension. We mitigate this through explicit documentation and by making the configuration layer programmable, but a formal Delphi study to validate benchmark consensus would strengthen the framework’s measurement reliability [13].

7 Discussion

7.1 Implications for Practice

The finding that Score Benchmark correlates weakly with commit count ($r = 0.36$) but strongly with story-point complexity ($r = 0.85$) has a practical implication: engineering managers who track commit frequency as a productivity proxy are likely missing a large portion of meaningful productivity variance. The FlowPMO Score Benchmark, by incorporating Review Quality, Process Conformance, and Anti-Rework, surfaces behaviors that commit-counting cannot capture.

The role-stratified analysis (RQ3) reveals a structural trade-off: Tech Leads exchange Delivery throughput and Flow Efficiency for Review Quality and Anti-Rework contributions. This is consistent with the expected role of a Tech Lead in an Agile team — they invest time in code review and mentoring rather than in individual delivery — but it means that applying the same absolute delivery benchmark to Tech Leads and individual contributors will systematically undervalue Tech Lead contributions. Practitioners using FlowPMO should consider role-specific benchmark profiles, or at minimum, interpret the Delivery dimension score for Tech Leads in the context of their review participation.

7.2 Absolute vs. Relative Normalization

The rank-collapse problem motivates absolute normalization, but absolute benchmarks introduce a different risk: benchmark staleness. A benchmark of 80% for Flow Efficiency may have been appropriate for teams using Scrum with two-week sprints, but may be unrealistically high for teams in domains with long approval cycles. We recommend that organizations using FlowPMO review benchmark values at least annually and after significant process changes.

7.3 Limitations of Euclidean Distance

The euclidean-distance formulation of Score Benchmark treats all five dimensions as equally important, which may not reflect organizational priorities. A weighted variant of Equation 2, where each dimension receives a weight w_i with $\sum w_i = 1$, would allow organizations to emphasize, for example, Anti-Rework over Delivery in a quality-critical domain. We leave the empirical validation of weighting schemes to future work.

8 Conclusion

This paper presented FlowPMO, an open-source developer productivity dashboard that operationalizes five SPACE-inspired dimensions using absolute benchmark normalization and a euclidean-distance composite index (Score Benchmark). We evaluated the framework on a reproducible synthetic dataset of 28 developers across 5 teams for Q3 2024.

Key findings are:

1. Score Benchmark captures productivity variance invisible to commit counts alone (Pearson $r = 0.36$ vs. $r = 0.85$ for story-point complexity score), validating its multi-dimensional design (**RQ1**).
2. The Delivery dimension exhibits the widest spread across developers (coefficient of variation: 39%), followed by Flow Efficiency (20%), confirming that throughput- and flow-related metrics are the most heterogeneous within the team (**RQ2**).
3. Tech Leads exhibit systematically higher Review Quality (+3.8 pp normalized) and Anti-Rework (+0.5 pp) but lower Delivery (−33.7 pp) and Flow Efficiency (−10.1 pp) compared to individual contributors, consistent with their process-coordination role (**RQ3**).

Future work will focus on: (i) validating the framework with longitudinal, multi-organization data; (ii) empirically calibrating dimension weights via a Delphi study with engineering managers; (iii) extending the framework to team-level aggregation with inter-dimensional correlation analysis; and (iv) integrating well-being and satisfaction proxies from pulse surveys to address the Satisfaction dimension of the SPACE framework.

References

- [1] T. DeMarco and T. Lister, *Peopleware: Productive Projects and Teams*, 2nd ed. Dorset House, 1999.

- [2] N. Forsgren et al., “The SPACE of Developer Productivity,” *ACM Queue*, vol. 19, no. 1, pp. 20–48, 2021.
- [3] B. Kitchenham, S. L. Pfleeger, and N. Fenton, “Towards a Framework for Software Measurement Validation,” *IEEE TSE*, vol. 21, no. 12, pp. 929–944, 1995.
- [4] D. J. Anderson, *Kanban: Successful Evolutionary Change for Your Technology Business*. Blue Hole Press, 2010.
- [5] S. M. A. Shah et al., “An Empirical Study on the Relationship Between Rework and Technical Debt,” in *Proc. ICSME*, 2023.
- [6] J. Caldeira et al., “Conformance Quality in Software Development: A Process Mining Approach,” in *Proc. ICPM*, 2019, pp. 121–128.
- [7] N. Fenton and S. L. Pfleeger, *Software Metrics: A Rigorous and Practical Approach*, 2nd ed. PWS Publishing, 1997.
- [8] M. Jørgensen, “What We Know About Software Development Effort Estimation,” *IEEE Software*, vol. 31, no. 2, pp. 37–40, 2014.
- [9] N. Forsgren, J. Humble, and G. Kim, *Accelerate*. IT Revolution Press, 2018.
- [10] W. M. P. van der Aalst, *Process Mining: Data Science in Action*, 2nd ed. Springer, 2016.
- [11] J. Caldeira et al., “Mining Software Development Rework Patterns Using Process Mining Techniques,” *IST*, vol. 135, p. 106565, 2021.
- [12] A. F. Nogueira and M. Z. Rela, “Continuous Integration Pipeline Quality,” *IST*, vol. 161, p. 107256, 2023.
- [13] B. Kitchenham, “Procedures for Performing Systematic Reviews,” Keele University, Tech. Rep. TR/SE-0401, 2002.